

MARSHFIELD, WI, USA

WMO: 726574

Lat: 44.638N

Lon: 90.188W

Elev: 383

StdP: 96.81

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94985

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-22.3	-29.1	0.3	-24.3	-26.5	0.4	-21.8	11.4	-10.5	10.4	-9.6	3.6	280	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	30.8	22.7	28.9	21.5	27.5	20.7	24.0	28.7	22.9	27.4	21.9	26.0	4.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.0	26.8	21.4	16.8	25.7	20.4	15.8	24.8	74.0	28.8	69.7	27.6	65.8	26.1	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.9	8.0	-28.3	33.3	3.5	2.3	-30.8	34.9	-32.8	36.2	-34.8	37.5	-37.3	39.1		
(5)			-28.5	25.9	3.5	1.3	-30.9	26.8	-33.0	27.6	-34.9	28.3	-37.4	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-9.6	-7.4	-0.7	6.7	13.4	18.6	21.0	19.8	15.8	8.5	0.9	-5.9		
	DBStd	11.89	7.01	6.73	6.73	5.34	4.62	3.75	3.10	3.09	4.44	5.00	5.71	6.32		
	HDD10.0	2442	608	487	336	124	19	1	0	0	6	91	277	493		
	HDD18.3	4443	867	720	589	350	165	42	10	19	100	307	523	752		
	CDD10.0	1286	0	0	6	26	123	258	342	303	180	44	4	0		
	CDD18.3	245	0	0	0	1	11	49	93	64	24	2	0	0		
	CDH23.3	2057	0	0	1	19	125	413	794	494	194	16	0	0		
	CDH26.7	519	0	0	0	3	28	104	228	115	41	1	0	0		
(14) Wind	WSAvg	3.8	4.1	4.1	4.0	4.4	4.1	3.4	3.1	2.9	3.3	3.9	4.0	4.0		
(15) Precipitation	PrecAvg	851	25	22	49	76	108	110	97	103	109	65	54	32		
	PrecMax	1165	68	75	120	146	264	243	267	245	270	143	146	93		
	PrecMin	548	2	0	2	12	25	18	13	18	8	7	0	7		
	PrecStd	138	16	17	28	35	55	61	53	51	63	32	34	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	9.8	21.1	26.2	29.7	32.4	33.7	32.0	30.3	25.9	17.9	11.2		
		MCWB	3.4	6.1	15.3	16.2	20.2	23.8	24.6	23.5	21.8	18.4	11.8	9.5		
	2%	DB	3.2	6.1	15.9	21.6	26.8	29.1	31.1	29.6	27.6	22.3	14.1	6.3		
		MCWB	1.4	3.4	11.0	13.0	18.3	21.5	23.2	22.2	20.3	16.4	10.3	3.9		
	5%	DB	1.3	3.4	12.1	18.5	24.0	27.5	29.0	27.6	25.8	19.2	11.7	3.7		
		MCWB	0.3	1.6	8.4	11.2	16.6	20.6	22.0	21.0	19.3	14.5	8.0	2.0		
	10%	DB	0.0	1.7	8.5	15.9	21.9	25.9	27.5	26.2	23.4	16.8	9.0	1.9		
		MCWB	-0.9	0.2	5.2	9.4	15.2	19.3	21.0	19.8	18.0	12.8	6.2	0.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	6.5	15.6	17.3	22.2	24.6	26.1	25.0	22.9	20.6	13.6	9.4		
		MCDB	5.8	9.5	20.2	23.8	27.8	29.8	32.0	29.4	27.9	23.1	16.3	10.8		
	2%	WB	1.7	3.5	12.2	14.4	19.8	23.1	24.3	23.4	21.6	17.4	11.0	4.4		
		MCDB	3.2	5.6	14.7	20.4	24.6	27.9	29.2	27.8	25.9	20.4	14.0	5.7		
	5%	WB	0.4	1.8	8.3	12.1	18.0	21.7	23.2	22.2	20.4	15.4	8.2	2.1		
		MCDB	1.3	3.4	12.1	16.4	22.4	26.1	27.6	26.2	23.9	18.8	10.8	3.4		
	10%	WB	-0.9	0.2	5.4	10.0	16.3	20.3	22.0	21.1	19.0	13.1	6.3	0.6		
		MCDB	-0.1	1.5	8.5	15.4	20.3	24.4	26.0	24.7	22.6	16.4	8.9	1.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.9	9.5	11.0	11.4	10.8	11.1	11.0	11.2	10.0	8.1	7.3		
		MCDBR	8.9	9.7	13.1	16.1	14.0	12.7	12.3	12.3	13.3	13.3	12.2	9.3		
		MCWBR	7.6	7.5	8.6	9.3	7.2	6.7	6.0	6.1	6.9	8.2	8.4	7.5		
		MCDBR	8.6	9.0	12.2	14.1	12.5	11.4	11.0	10.9	11.3	11.9	11.1	8.3		
	5% WB	MCWBR	7.5	7.3	8.4	9.0	7.1	6.7	6.2	6.1	6.7	8.3	8.5	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.273	0.311	0.346	0.379	0.380	0.374	0.378	0.349	0.316	0.285	0.257		
	taud		2.409	2.335	2.387	2.379	2.330	2.355	2.382	2.369	2.452	2.529	2.529	2.456		
	Ebn at Noon		893	922	926	916	890	888	889	873	875	865	842	844		
	Edh at Noon		71	94	103	114	123	121	116	114	96	78	65	63		
(44) All-Sky Solar Radiation	RadAvg		1.71	2.84	3.95	4.53	5.39	5.92	6.13	5.19	4.10	2.60	1.65	1.23		
	RadStd		0.19	0.26	0.44	0.45	0.41	0.46	0.30	0.35	0.28	0.30	0.16	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page